Database History
(Greatly Abridged and Simplified)

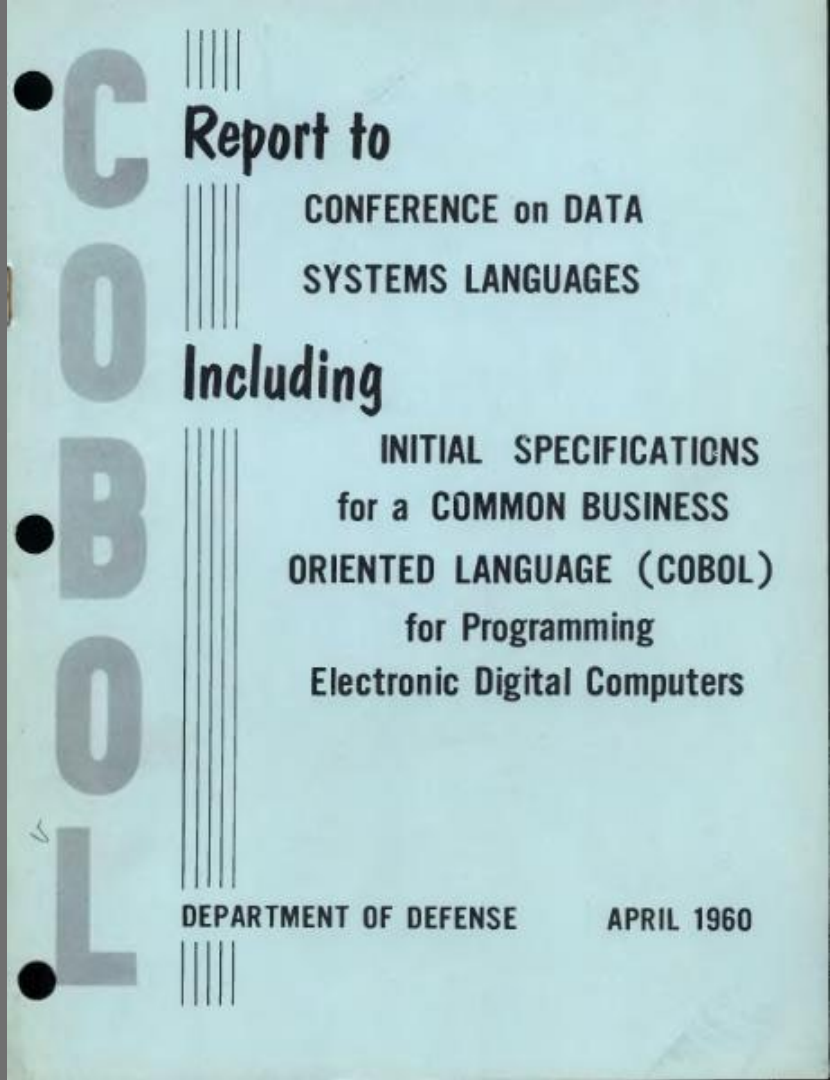
ANCIENT HISTORY: 1950’S-1970’S

Computing in the 1950’s

Data management almost inconceivable.

CODASYL (1959)

First COnference on DAta SYstems and Languages



COBOL Sample (Part 1)

```
000010 IDENTIFICATION DIVISION.
000020 PROGRAM-ID. SAMPLE.
000030 AUTHOR. J.P.E. HODGSON.
000040 DATE-WRITTEN. 4 February 2000
000041
000042* A sample program just to show the form.
000043* The program copies its input to the output,
000044* and counts the number of records.
000045* At the end this number is printed.
000046
000050 ENVIRONMENT DIVISION.
000060 INPUT-OUTPUT SECTION.
000070 FILE-CONTROL.
000080     SELECT STUDENT-FILE ASSIGN TO SYSIN
000090         ORGANIZATION IS LINE SEQUENTIAL.
000100     SELECT PRINT-FILE ASSIGN TO SYSOUT
000110         ORGANIZATION IS LINE SEQUENTIAL.
000120
000130 DATA DIVISION.
000140 FILE SECTION.
000150 FD  STUDENT-FILE
000160     RECORD CONTAINS 43 CHARACTERS
000170     DATA RECORD IS STUDENT-IN.
000180 01  STUDENT-IN              PIC X(43).
000190
000200 FD  PRINT-FILE
000210     RECORD CONTAINS 80 CHARACTERS
000220     DATA RECORD IS PRINT-LINE.
000230 01  PRINT-LINE              PIC X(80).
000240
```

```
000250 WORKING-STORAGE SECTION.
000260 01  DATA-REMAINS-SWITCH     PIC X(2)
000261                 VALUE SPACES.
000262 01  RECORDS-WRITTEN         PIC 99.
000270
000280 01  DETAIL-LINE.
000290     05 FILLER               PIC X(7)
000291                 VALUE SPACES.
000300     05 RECORD-IMAGE         PIC X(43).
000310     05 FILLER               PIC X(30)
000311                 VALUE SPACES.
000312
000313 01 SUMMARY-LINE.
000314     05 FILLER               PIC X(7)
000315                 VALUE SPACES.
000316     05 TOTAL-READ           PIC 99.
000317     05 FILLER               PIC X
000318                 VALUE SPACE.
000319     05 FILLER               PIC X(17)
000320                 VALUE 'Records were read'.
000321     05 FILLER               PIC X(53)
000322                 VALUE SPACES.
000323
```

Data described declaratively, separate from “code”.
Data arranged in records.

COBOL Sample (Part 2)

```
000330 PROCEDURE DIVISION.
000331
000340 PREPARE-SENIOR-REPORT.
000350     OPEN INPUT STUDENT-FILE
000360          OUTPUT PRINT-FILE.
000361     MOVE ZERO TO RECORDS-WRITTEN.
000370     READ STUDENT-FILE
000380         AT END MOVE 'NO' TO DATA-REMAINS-SWITCH
000390     END-READ.
000400     PERFORM PROCESS-RECORDS
000410         UNTIL DATA-REMAINS-SWITCH = 'NO'.
000411     PERFORM PRINT-SUMMARY.
000420     CLOSE STUDENT-FILE
000430         PRINT-FILE.
000440     STOP RUN.
000450
000460 PROCESS-RECORDS.
000470     MOVE STUDENT-IN TO RECORD-IMAGE.
000480     MOVE DETAIL-LINE TO PRINT-LINE.
000490     WRITE PRINT-LINE.
000500     ADD 1 TO RECORDS-WRITTEN.
000510     READ STUDENT-FILE
000520         AT END MOVE 'NO' TO DATA-REMAINS-SWITCH
000530     END-READ.
000540
000550 PRINT-SUMMARY.
000560     MOVE RECORDS-WRITTEN TO TOTAL-READ.
000570     MOVE SUMMARY-LINE TO PRINT-LINE.
000571     WRITE PRINT-LINE.
000572
```

Code processes a record at a time.
English-like syntax.
Keywords capitalized.

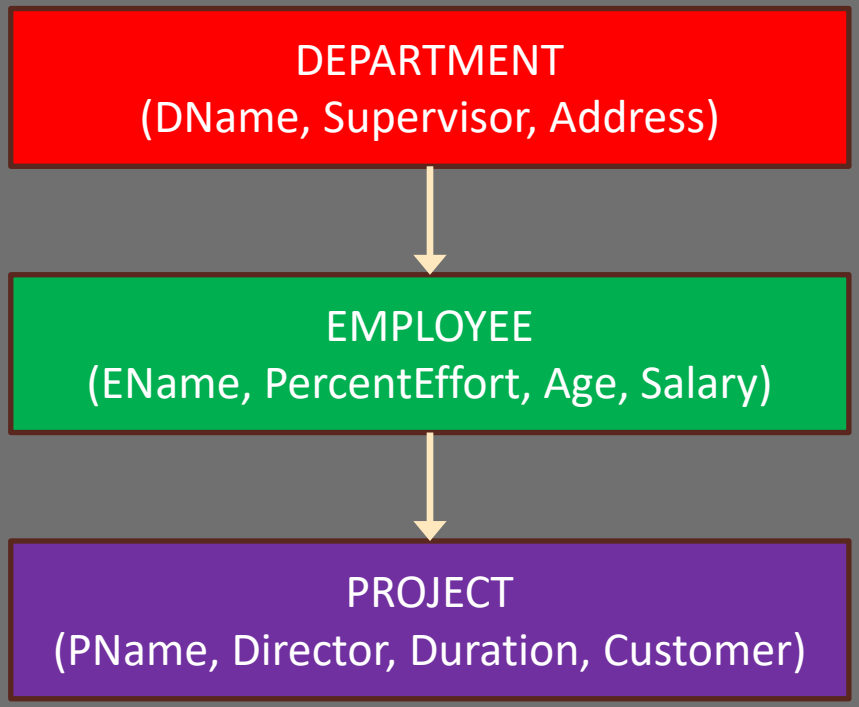
Information Management System (1966)

Developed by IBM for Saturn V

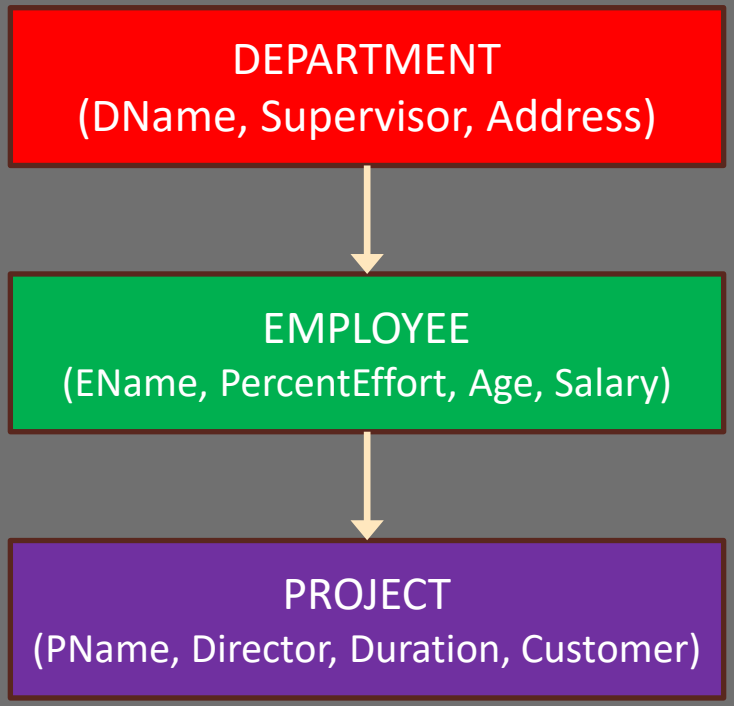
Used first serious data model – hierarchical



Hierarchical Data Model



Hierarchical Data Model

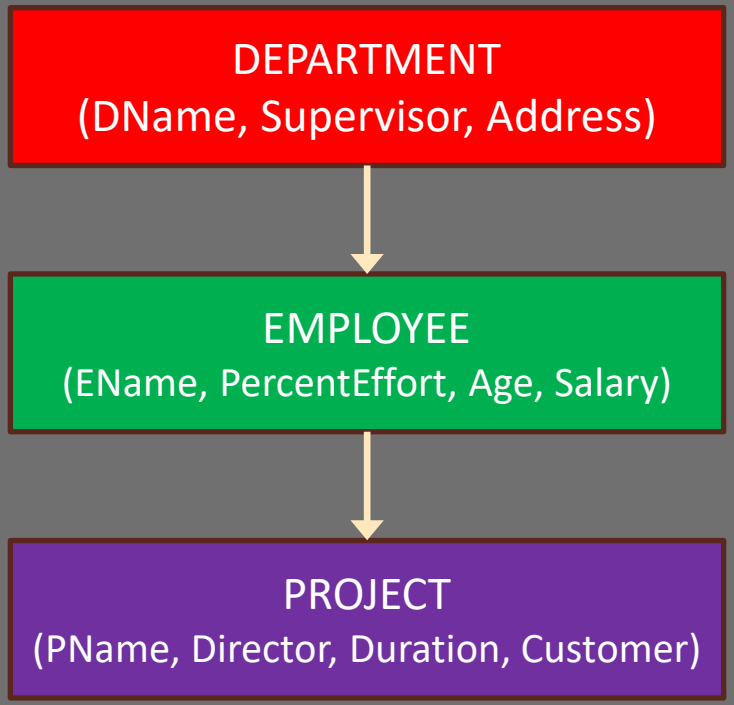


Which employees supervised by Mary work on Fox?

DL/1 code:

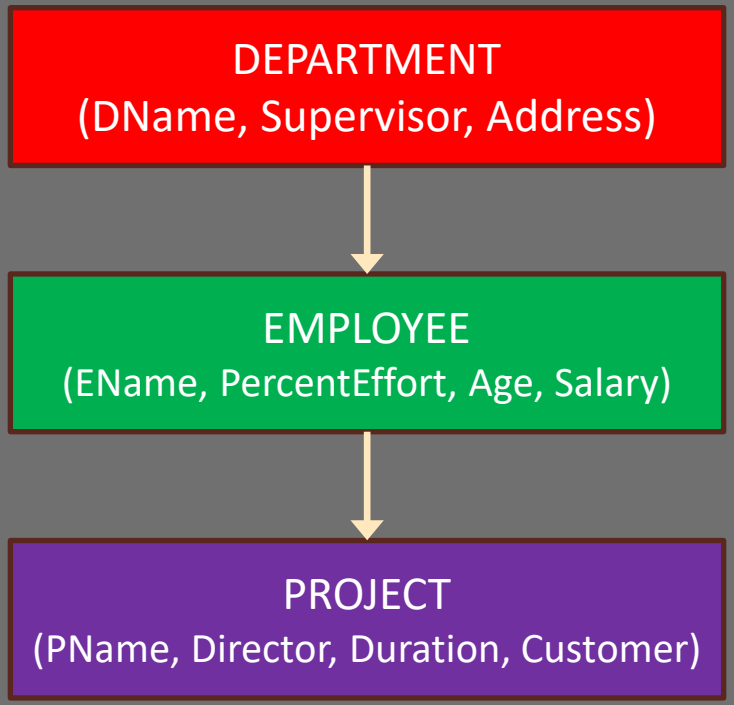
```
Get unique Department (Supervisor = "Mary")
  Until failure do
    Get next within parent
      Until failure do
        Get next within parent (PName = "Fox")
        ...
      Enddo
  Enddo
```

Hierarchical Data Model – directional



Can search from Departments to Projects, but not the other way.

Hierarchical Data Model – redundancy

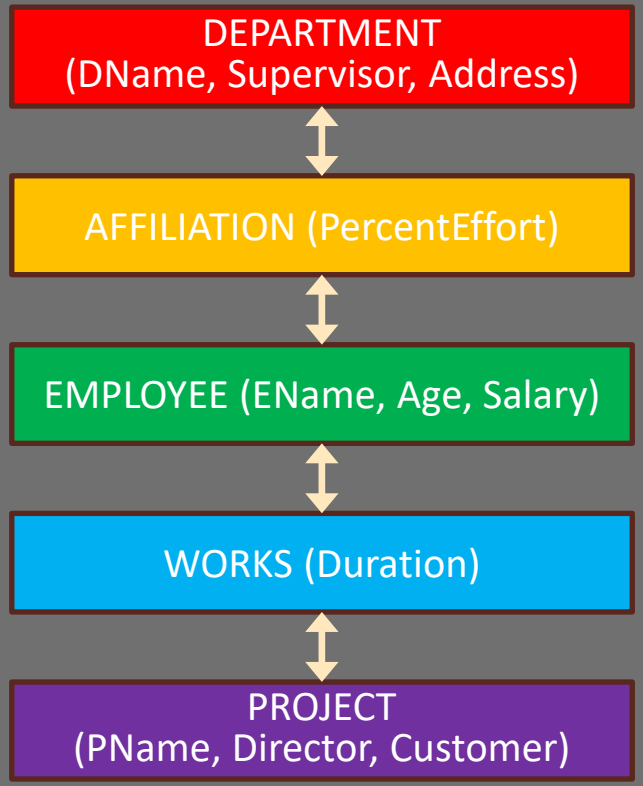


Assume:

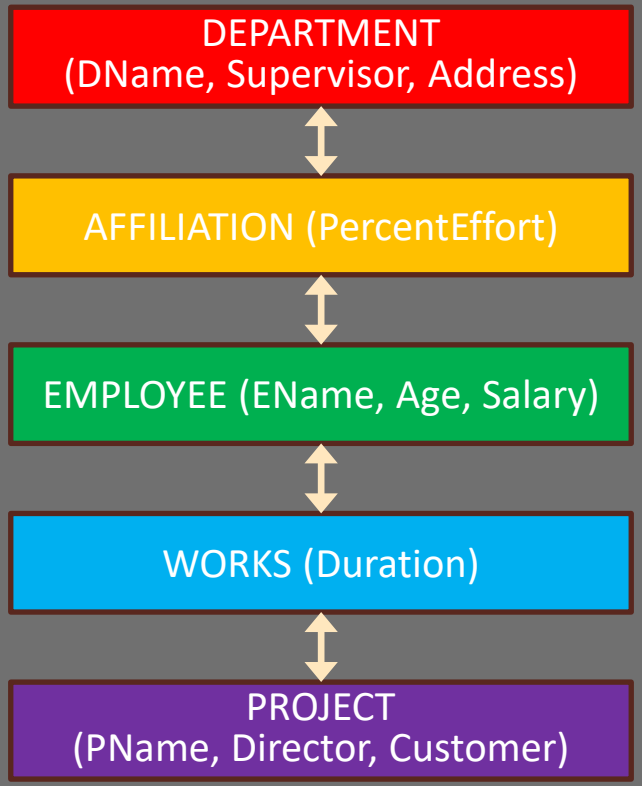
- Employees can work on multiple projects.
- Projects can have multiple employees.

Redundancy: Each employee repeats project info.

CODASYL Network Data Model (1969)



CODASYL Network Data Model (1969)

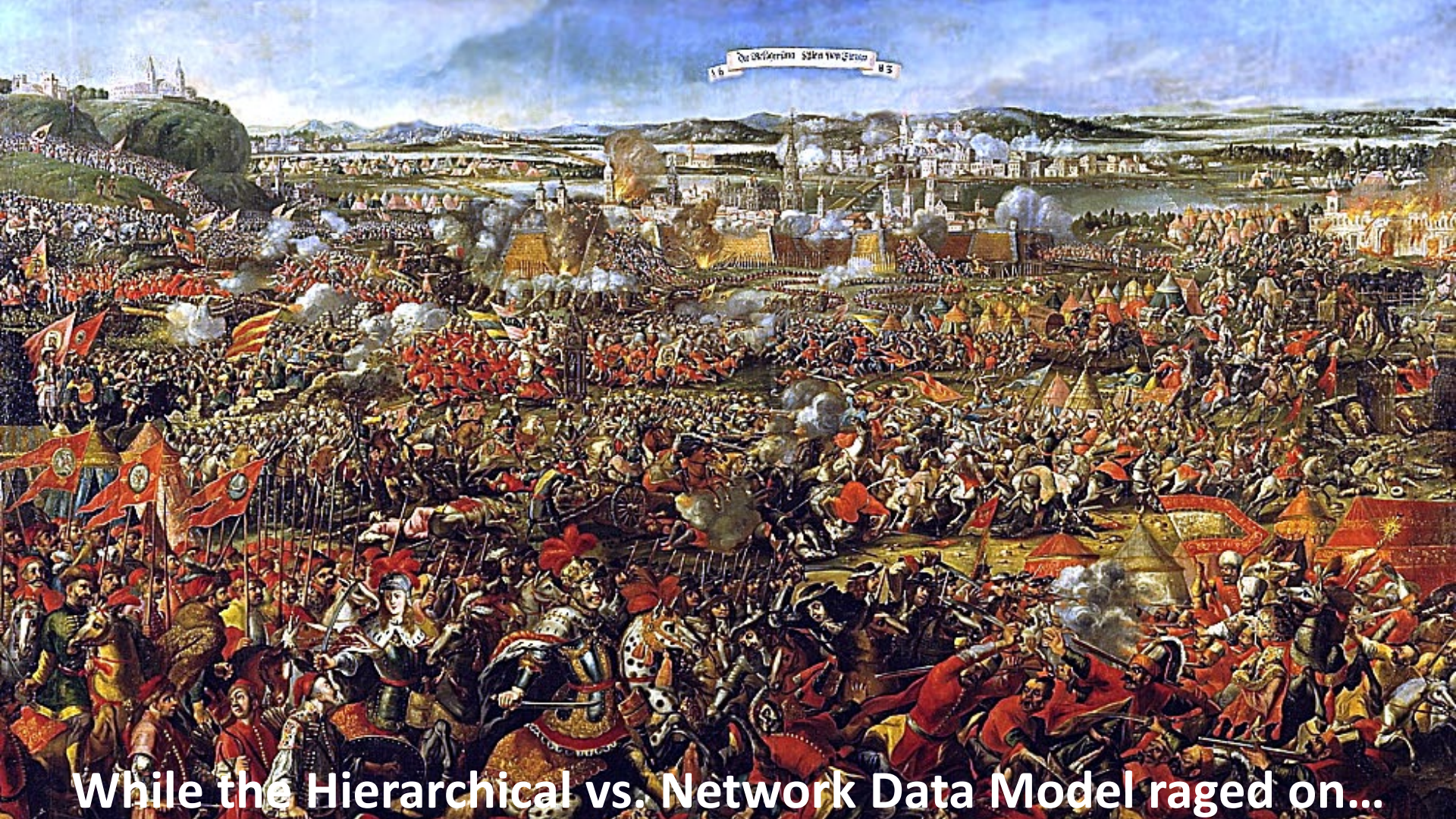


Which employees supervised by Mary work on Fox?

Search ↑ good when few on project Fox.

Search ↓ good when few supervised by Mary.

However, code searches in one direction.


While the Hierarchical vs. Network Data Model raged on…

Relational Model (1970)

DEPARTMENT
(DName, Supervisor, Address)

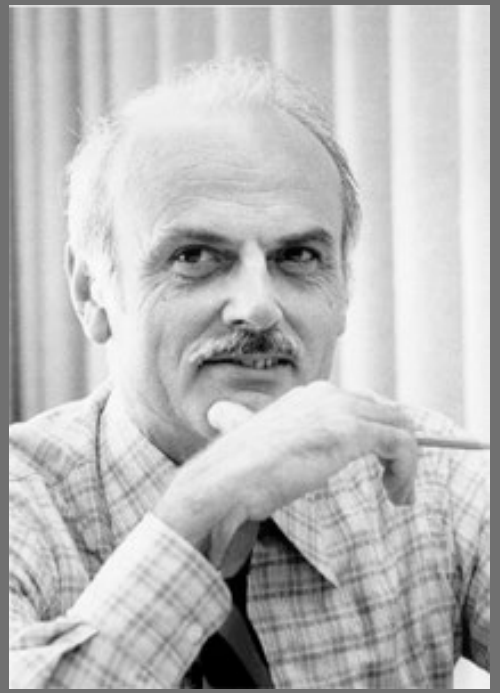
AFFILIATED WITH
(DName, EName, PercentEffort)

EMPLOYEE (EName, Age, Salary)

WORKSFOR
(Ename, Pname, Duration)

PROJECT
(PName, Director, Customer)

Edgar F Codd:
"A relational model of data for large shared data banks."



Relational Model

DEPARTMENT
(DName, Supervisor, Address)

AFFILIATED WITH
(DName, EName, PercentEffort)

EMPLOYEE (EName, Age, Salary)

WORKSFOR
(Ename, Pname, Duration)

PROJECT
(PName, Director, Customer)

Which employees supervised by Mary work on Fox?

```
{e.EName | Employee(e) and
           exists (w,a,d) (WorksFor(w) and
                           AffiliatedWith(a) and
                           Department(d) and
                           e.EName = w.EName and
                           w.PName = "Fox" and
                           a.EName = e.Ename and
                           a.DName = d.DName and
                           d.Supervisor = "Mary") }
```

MODERN HISTORY: 1980’S-PRESENT

Object-Oriented DBs (1980s)

Persistent objects



Then: Didn't catch on.

- Simple objects fit well into relational model.
- Lost advantages of declarative queries. Largely repeats hierarchical & network models.
- Different DBMS for each language.
- Wasn't fast enough.

VERSANT

Now: Idea has returned with more complex objects.

- Same library for multiple languages.
- Same data representation in application & DB.



MapReduce, Spark, Hadoop

Not a DB. Frameworks for analyzing large data sets.

Highly publicized non-SQL approach

- Uses first-order functions – fits well with modern PLs
- Easily parallelized







NoSQL

Types

- Document store
- Key/value stores
- Column stores
- Graph DBs

Benefits

- Flexible schema
- More easily scalable
- Faster for what each type is designed for